

RESTRICTED

Code No. _____

THE INFORMATION GIVEN IN THIS DOCUMENT IS NOT TO BE PUBLISHED OR COMMUNICATED, EITHER DIRECTLY OR INDIRECTLY TO THE PRESS OR ANY PERSON NOT HOLDING AN OFFICIAL POSITION IN THE SERVICE OF THE GOVERNMENT OF INDIA



Technical Manual

Part I, Technical Info
Volume 1 of 2, Technical Description

Ruggedized Field Telephone Set (RFT1001)

Doc ID: 1551/1-DPAR002014

THIS HANDBOOK DOES NOT CONSTITUTE AUTHORITY FOR REPAIRS OR REPLACEMENT OF COMPONENTS BY USERS. REPAIRS BY DIFFERENT AGENCIES WILL ONLY BE CARRIED OUT AS LAID DOWN IN PERMISSIBLE REPAIR SCHEDULES PUBLISHED AS EMERs OR EQUIVALENT FROM TIME TO TIME.

Prepared and Published by:

Elcom Innovations Pvt. Ltd.
B-76, Phase 7, Industrial Area, Sector 74,
SAS Nagar, Punjab 160055, (India)

April, 2026

Under the Authority of,
Secretary, Ministry of Defence, Govt of India

RESTRICTED

RECORD OF CHANGES

Change Notice No.	Authority	Incorporated by “Name and Rank” in Block letters	Initials with date
(1)	(2)	(3)	(4)

ASSOCIATED HANDBOOKS

Technical Literature for **FIELD TELEPHONE SET RFT1001**
comprises of the followings:

- USER HANDBOOK
- TECHNICAL MANUAL

PART I TECHNICAL INFORMATION

Volume 1 **Technical Description**

Volume 2 Drawings

PART II MAINTENANCE

PART III OVERHAUL AND RECONDITIONING INSTRUCTIONS

PART IV MANUFACTURER'S PARTS LIST

Volume 1 Parts List

Volume 2 Illustrations

SAFETY WARNING

**The voltages employed in this Equipment are sufficiently
high to endanger human life.**

**POWER
MUST BE SWITCHED OFF
before servicing the equipment
&
GREAT CARE
taken when making internal adjustments etc.**

FOR FIRST AID IN CASE OF ELECTRIC SHOCK SEE OVERLEAF

FIRST AID IN CASE OF ELECTRIC SHOCK

FIG. 1.



FIG. 2.



FIG. 3.



FIG. 4.



1.	SWITCH OFF. If this is not possible, PROTECT YOURSELF with dry insulating material and pull the victim clear of the conductor. DO NOT TOUCH THE VICTIM WITH YOUR BARE HANDS until he is clear of the conductor, but DO NOT WASTE TIME.
2.	(a) Place the victim in the supine position. (b) Keep the air passages clear by turning the head to one side, opening the patient's mouth and clearing it of water, saline, mucus or blood, a lot of which might have accumulated in the back of the throat.(Figure 1).
3.	If the jaw is rigid try to force the mouth open by pressure on the gum behind the last molar tooth of the lower jaw. When the upper air passages are thus cleared, tilt the head backward and force the jaw forwards from the angles of the jaw in front of the ears. This would prevent mechanical obstructions to the upper air passages.(Figure 2-3)
4.	(a) Then hold the chin up and forward with one hand and pinch the nostrils of the victim with the other.(Figure 4) (b) Take a very deep breath and apply your mouth to that of the victim and blow into his mouth, until the chest of the victim moves up indicating filling of the lungs. (NEVER ALLOW THE CHIN TO SAG).(Figure 4)
5.	When the chest has moved up, withdraw your mouth and allow the chest to sink back. REPEAT this process every three to four seconds until the victim begins to breathe again or until he is taken over by a medical attendant. This method can be continued in an ambulance during transit of the patient from the site of accident to the nearest medical centre.
HAVE SOMEONE ELSE AND SEND FOR A DOCTOR. KEEP PATIENT WARM AND LOOSEN HIS CLOTHING. DO NOT GIVE LIQUIDS UNTIL THE PATIENT IS CONSCIOUS.	

CAUTION/NOTES

Ringer output Voltage in this equipment is sufficiently high to endanger human life. Powers of external battery connections / internal batteries are to be removed before servicing the equipment.

CAUTION

Do not touch the exposed metal conductor of the field wire, when the phone is in Magneto mode and Magneto Call button is pressed.

DEMOLITION OF MATERIAL TO PREVENT ENEMY USE

(A) General

The demolition procedure outlined below will be used to prevent the enemy from using or salvaging this equipment. **DEMOLITION OF THE EQUIPMENT WILL BE DONE ONLY ON ORDER OF THE COMMANDER.**

(B) Methods of Destruction

Use any or all the following methods to destroy the equipment.

- (i) Smash - Smash the crystals, controls semiconductors, coil, switches, transformers and handset. Use sledges, hand axes, pickaxes, hammers crowbars and heavy tools.
- (ii) Cut - Cut the cords, headsets and wiring. Use axes, hands axes or hatchets.
- (iii) Burn - Burn the cords, resistors, capacitors, coils, wiring, technical manuals. Use gasoline, kerosene, oil, flamethrowers or incendiary.
- (iv) Bend - Bend the panels, cabinet and chassis.
- (v) Explosives - If explosives are necessary, use firearms, grenades or TNT.
- (vi) Disposal - Bury or scatter the destroyed parts in slit trenches, fox holes or throw them into streams.

CHANGE NOTICE

TITLE OF PUBLICATION	CODE NO. AND DATE	PREVIOUS CHANGE	CHANGE NOTICE NO. AND DATE

No. of Pages:
Enclosures:
Sheets:

CODES FOR INCORPORATION OF CHANGES	NOTE
(R) By replacement / addition of sheet(s)	This document should be filled / pasted in the copy of the publication before the CONTENTS page. After carrying out (R), (H) or (P), entry should be made in Record of changes Page.
(H) By Hand – correction	
(P) By pasting enclosed matter	
(A) Additional information – only to be filled in publication	

Note:

Table of Contents

1. Scope	17
1.1 Reference Document.....	17
1.2 Intended Audience	17
1.3 Abbreviations	17
2. About The Product	19
2.1 Salient Features	19
3. Technical Specifications of Field Telephone RFT1001.....	21
3.1 Mechanical Specifications	21
3.2 Electrical Specifications.....	21
3.3 Environmental Characteristic.....	23
4. Principles of Operation	24
4.1 Working Principle	24
4.2 Mode of Operation.....	24
4.2.1 Auto Mode (CBT / CBP)	24
4.2.2 Magneto Mode (LB)	25
4.2.3 RAD Mode - Magneto Mode with CNR Radio	26
4.3 Functional Description.....	27
4.4 Description of Controls, Terminals & LEDs	31
4.4.1 Line Terminals	31
4.4.2 External Battery Terminals	31
4.4.3 Ringer Volume Switch - RINGER.....	31
4.4.4 Receive Volume Switch - SPEECH.....	31
4.4.5 Pulse/Tone Indicator – P/T.....	31
4.4.6 Ring Voltage Indicator - RV	32
4.4.7 Battery Monitoring Key – BAT MON.....	32
4.4.8 Battery Status Indicators (G & P)	32
4.4.9 Reverse Battery Indicator – R.....	32
4.4.10 MAGNETO CALL Button.....	32
4.4.11 Dial Keypad	33
4.4.12 SYSTEM Switch.....	33
4.4.13 OFF/ON Hook Switch.....	33
4.4.14 PTT Switch.....	33
4.4.15 Internal Battery Box.....	33
5. Detail Circuit Theory	34
5.1 EBA30058, Control Card	35
5.1.1 Speech Circuit, Line Loop Circuit & RC Coil	35
5.1.2 Amplifier Circuit.....	36
5.1.3 Dial Circuit.....	36
5.1.4 Keypad Circuit	36
5.1.5 BAT. MON. & Ring Indicator Circuit	36
5.1.6 Connectors Detail.....	36
5.1.7 Handset Interface & PTT	39

5.2 EBA30059, Ringer Card PCA	40
5.2.1 Connector Details	40
5.3 66079, Hook Switch Assembly	41
5.3.1 Connector Details	41
5.4 EBA30060, Ringer Micro Switch Card	42
5.4.1 Connector Details	42
5.5 EBA30062 System Switch Card PCA	43
5.5.1 Connector Details	44
5.6 EBA30063, BAT MON Switch Card	46
5.6.1 Connector Details	46
5.7 Ringer Volume Switch Assembly	47
5.7.1 Connector Details	47
5.8 TX/RX_VOL_(TRV) Volume Switch Assembly	48
5.8.1 Connector Details	49
5.9 Handset Connector Assembly	50
5.9.1 Connector Details	50
6. Document History	51

List of Figures

Figure 1: General View of Field Telephone, RFT1001	19
Figure 2: RFT1001 in CBT/CBP Mode	24
Figure 3: RFT1001 in LB Mode with Another LB Phone	25
Figure 4: RFT1001 in LB Mode with Magneto Exchange	25
Figure 5: RFT1001 in RAD Mode	26
Figure 6 : Block diagram of Field Telephone Set RFT1001	27
Figure 7: Controls & Terminals	29
Figure 8: SYSTEM Switch & Battery Box of RFT1001	29
Figure 9: Hook Switch & PTT	30
Figure 10: External & Internal Battery Connections	30
Figure 11 : Interconnection of Field Telephone Set RFT1001	34
Figure 12 : Block Diagram of EBA30058, Control Card	35
Figure 13: Block Diagram-EBA30059, Ringer Card	40
Figure 14: Block Diagram – 60079, Hook Switch Card	41
Figure 15: Block Diagram – EBA30060, Ringer Micro Switch Card	42
Figure 16: Block Diagram – EBA30062, System Switch Card	43
Figure 17: System Switch-Interconnection Diagram	43
Figure 18: Block Diagram of EBA30063, BAT MON Switch Card	46
Figure 19 : Volume Switch-Interconnection Diagram	47
Figure 20 : TX/RX Volume Switch-Interconnection Diagram	48
Figure 21: Block Diagram-Handset Connector	50

List of Tables

Table 1: Mechanical Specifications of RFT1001.....	21
Table 2: Electrical Specifications of RFT1001	22
Table 3: Environmental Characteristics of RFT1001	23
Table 4: Control, Terminals & Indicators of RFT1001	28
Table 5: P/T Indicator	31
Table 6: Battery Monitoring	32
Table 7: SYSTEM Switch.....	33
Table 8: Control Card PCA Connector Pin Details	38
Table 9: Handset Connector Details	39
Table 10: EBA30059 Ringer Card, Connector Pin Details.....	40
Table 11: Hook Switch Assembly - Cconnector Pin Details	41
Table 12: Ringer Micro Switch Card PCA - Connector Pin Details.....	42
Table 13: System Switch Card - Connector Pin Details	44
Table 14: Signal Status in Various Positions of System Switch.....	45
Table 15: EBA30063, BAT MON Switch Card, Connector Pin Details	46
Table 16: BZ-VOL_(BZV) Connector Pin Details.....	47
Table 17: Signal Status in Various Positions of BZ-Volume Switch	48
Table 18: Volume switch PCA Connector Pin Details	49
Table 19: Signal Status in Various Positions of TX/RX VOL (TRV) Switch.....	49
Table 20 : Handset Connector Assembly Pins Detail	50
Table 21 : Document History.....	51

Note:

1. Scope

- a) This document (1551/5-DPAR002014) is one of the technical document which covers part lists of RFT1001. This document shows the top level part list along with the part list of various sub-assemblies such as mechanical sub-assemblies, electronic cards assemblies, wire harnesses assemblies. The section 1.1 shows complete list of technical documents of RFT1001.
- b) Changes/additions which may have to be incorporated from time to time as supplement to this Technical Manual Part IV- Volumes 1 and 2 (to be called as Parts Lists) will be issued by **M/s. Elcom Innovations Pvt. Ltd** covering:
- Identification details of Sub-Assemblies which were not available at the time of preparations.
 - Allotment of DS/ Service Catalogue and Part nos. not included earlier.
 - Changes in the equipment due to modifications/ Updation.
- c) User should amend this book and record details of changes in the Record of Changes page.

1.1 Reference Document

[1] User Handbook	1553-DPAR002014
[2] Technical Manuals,	
PART I - Technical Information, Description (Vol 1 of 2)	1551/1-DPAR002014
Technical Information, Drawings (Vol 2 of 2)	1551/2-DPAR002014
PART II - Maintenance	1551/3-DPAR002014
PART III - Overhaul & Reconditioning Instructions	1551/4-DPAR002014
PART IV - Manufacturers' Parts List – Part Lists (Vol 1 of 2)	1551/5-DPAR002014
Manufacturers' Part List – Illustrations (Vol 1 of 2)	1551/6-DPAR002014

1.2 Intended Audience

The intended audience of this document is the trained technical personal on Field Telephone Set RFT1001.

1.3 Abbreviations

AC	Alternative current
CBT	Common battery with DTMF dialing
CBP	Common battery with Pulse dialing
BAT MON	Battery monitoring
DTMF	Dual Tone Multiple Frequency
DC	Direct current
EMI	Electromagnetic Interference
EMC	Electromagnetic compatible
IDP	Inter Digit Pause
LED	Light Emitting Diode
LB	Local battery
RAD	Local battery with radio set
PPS	Pulse Per Second
P/T	Pulse/Tone
PCA	Printed circuit assembly

PTT	Press to Talk
RH	Relative Humidity
RV	Ringer Visual
mA	milli ampere
mV	milli volt
MTBF	Mean time between failures
MTTR	Mean time to repair
Kg	Kilogram
mV	Millivolt

2. About The Product

The RFT1001 is a lightweight, ruggedized and portable field telephone. It can function as a normal magneto telephone (LB phone), capable of working with another magneto telephone of similar kind, or with a magneto exchange. It can also work as an auto telephone (CB phone) with dialling facility (either in PULSE or DTMF). The field telephone has a built-in dial keypad. In magneto phone mode, the field telephone works from 3 nos. of 1.5 Volt standard D type dry batteries.



Figure 1: General View of Field Telephone, RFT1001

2.1 Salient Features

- a) Compact, lightweight and ruggedized using casted aluminium alloy painted with olive green colour.
- b) Easy to install, operate and maintain.
- c) Operates from 3 nos. of 1.5V, 'D' type dry battery.
- d) Battery consumption under normal usage is minimal.
- e) Telephone is protected against reverse battery connections.
- f) The field telephone has a built-in dial keypad.
- g) The dial keypad is disabled during magneto mode (LB) and Radio mode (RAD).
- h) Handset of telephone uses Noise cancelling for improved speech communication in places with high ambient noises.
- i) The design of handset is user friendly and can be comfortably used while wearing helmet. The handset is held on telephone by use of spring-loaded clamps.
- j) For the easy maintenance purpose for a lineman, following visual indications are available in telephone set.
 - Visual indication when line is short.

- Visual indication for battery health.
 - Visual indication for incoming ring.
 - Visual indication for outgoing ring.
 - Visual indication for reversed polarity of battery connection
- k) Can be used with similar field telephone or any magneto telephone and magneto exchange.
- l) Can be used on auto exchange lines.
- m) Dial keypad supports both PULSE and DTMF dialling.
- n) Dynamic receiver and microphone for high quality performance.
- o) Visual ringing indicator for incoming ring with auto exchange mode.
- p) Visual ringing indicator for incoming and outgoing ring with CBT, CBP & LB mode.
- q) Push button operation for magneto calling.
- r) Ringer volume can be set to MUTE, LOW & HIGH by adjusting the SPEECH switch.
- s) Ringing tone is loud enough to be heard upto max 0f 3 Meters from telephone in ambient noise conditions.
- t) Line terminals and Battery terminals are spring loaded type for easy access but at the same time protected against damage due to accidental dropping of the telephone.
- u) Visual indication for battery status.
- v) Carrying pouch is suitably ruggedized as per military standards. It is olive green and has an adjustable strap for easy carrying. The telephone can also be operated without taking it out from the carrying pouch. The telephone can easily be removed from the case without the use of any special tools.
- w) The mechanical design of the telephone comprises the captive screws to avoid the loss of screws during field maintenance.

3. Technical Specifications of Field Telephone RFT1001

3.1 Mechanical Specifications

S/N	Specification	Range
1	Weight (In Kg)	a) 2.14 Kg with battery & handset b) 1.90 Kg with handset (w/o battery)
2	Weight of Handset	370gms
3	Dimension (in mm) Max	Height : 110 ± 1 mm Width : 124 ± 0.5 mm Length : 235.5 ± 1 mm

Table 1: Mechanical Specifications of RFT1001

3.2 Electrical Specifications

S/N	Specification	Range
1	Mode of Operation: a) CBT b) CBP c) LB d) RAD	a) Central Battery with DTMF dialing b) Central Battery with decadic dialing c) Local Battery (Magneto mode) d) To connect with CNR Radio through external device RAU (optional)
2	Power Specs- a) Internal Battery b) Max Current Drain c) Idle current d) Max speech current e) Ext battery voltage	a) 4.5 Vdc, 3x1.5V 'D' type dry battery b) 180 mA (max) c) Nil d) 16 mA e) 4.5 Volt DC
3	Internal Battery Monitoring	Good, Poor & Reverse *Reverse is only for internal batteries
4	Loop Resistance a) In CBT/CBP Mode b) In LB Mode c) In RAD Mode (Loop Test)	a) 800 Ohms (max.) line loop (with 48V line voltage) b) 7500 Ohms (max.) line loop. c) 360 to 440 Ohms across line terminals when PTT is pressed.
5	Speech Range	55 dB
6	Speech Output (La – Lb terminals)	More than 900 mV.
7	Side Tone	300 mV (min) below Transmit level (Speech Output).

S/N	Specification		Range
8	Receiver Frequency		0.3 Khz to 3.4 Khz
9	Ringing Voltage (LB Mode)		a) 90 V (minimum) at no Load b) More than 18V with 2KΩ Load
10	Receive Efficiency		Not less than 350 mV
11	Send efficiency		Not less than 900 mV
	Freq Response	300 Hz	Should not less than 450 mV
		500 Hz	
		2000 Hz	
		3000 Hz	
		3400 Hz	
	Side Tone Efficiency		At least 300 mV below the value of speech output test.
12	Signaling characteristics a) In CBT & CBP b) In LB		a) PPS : 10 ± 0.5 IDP : 800 ± 10% Flash : 300/600 ± 20 msec Make/Break : 60 % – 70 % b) Ring Freq : 25 Hz ± 2 Hz
13	Insulation resistance		Better than 5 MΩ at 250 V DC across body and line terminals
14	Lineman test facility		LED indication for following condition a) Line Open b) Line Short

Table 2: Electrical Specifications of RFT1001

3.3 Environmental Characteristic

1	Operating temperature	-25°C to +55°C
2	Storage temperature	-40°C to +65°C
3	Relative Humidity	0 – 95 % below +30°C
4	High altitude	Upto 5000 meter above sea level at -20°C
5	Driving rain test	100 mm/hour
6	Immersion	Upto 1 meter for 2 hour.

Table 3: Environmental Characteristics of RFT1001

4. Principles of Operation

4.1 Working Principle

The Field Telephone RFT1001 is a rugged, lightweight and waterproof analogue telephone set designed according to defence requirements and suitable for connection to manual (magneto) or automatic 2-wire switchboards, to a similar telephone or directly to the public telephone network (PSTN).

The telephone electronics use a high efficiency electronic ring generator to ensure low power consumption and long battery life. The Field Telephone RFT1001 operates in Common Battery (CB) and Local Battery (LB) modes, as selected by the user. When operating in LB modes, the Field Telephone Set RFT1001 shall be powered by three batteries of 'D' size. Batteries are not required in CB mode.

A slim and lightweight military standard handset with push-to-talk switch shall allow easy operation even by personnel wearing helmets.

4.2 Mode of Operation

Field Telephone Set RFT1001 operates in the following modes

4.2.1 Auto Mode (CBT / CBP)

Figure 2 shows the connectivity diagram to install the field telephone in **CBT** or **CBP** mode. In auto mode (CBT/CBP), the telephone is powered by common battery exchange and does not use its internal batteries.

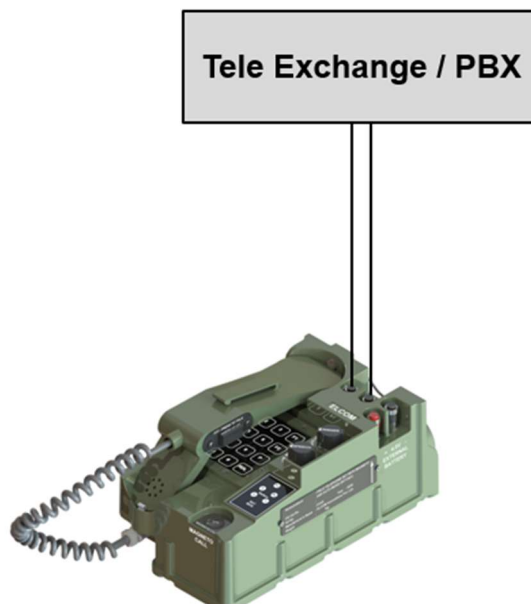


Figure 2: RFT1001 in CBT/CBP Mode

4.2.2 Magneto Mode (LB)

As described earlier, the field telephone can function as a normal magneto telephone with a telephone of similar kind, or with a magneto exchange.

Figure 3 and Figure 4 shows the connectivity diagram to install the field telephone in LB mode with another magneto telephone and another magneto exchange respectively.

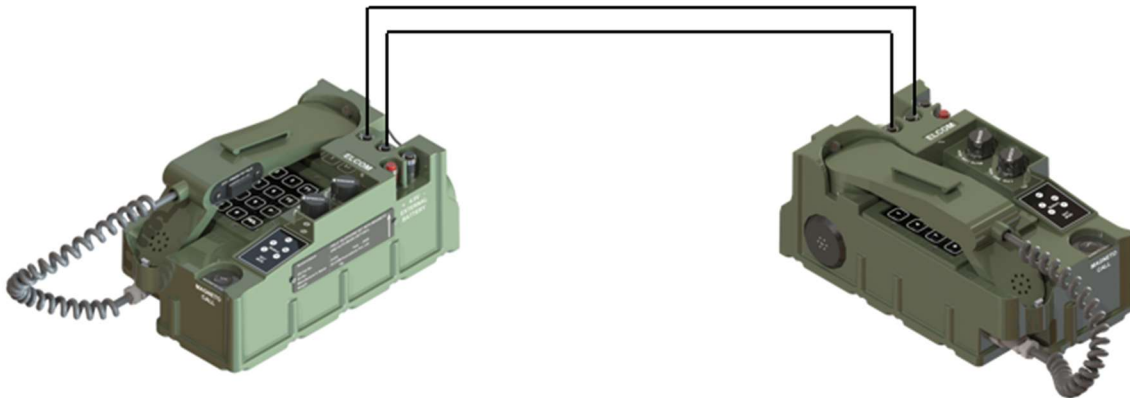


Figure 3: RFT1001 in LB Mode with Another LB Phone

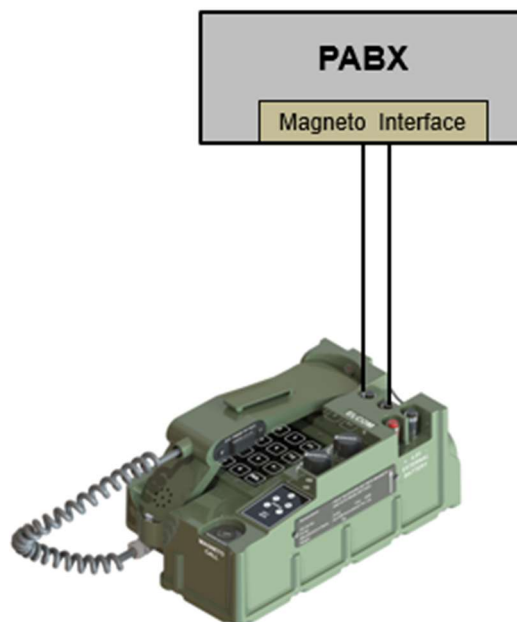


Figure 4: RFT1001 in LB Mode with Magneto Exchange

4.2.3 RAD Mode - Magneto Mode with CNR Radio

The field telephone RFT1001 shall be used in **RAD** mode if the phone has to be interfaced with a CNR radio set through an external Radio Interface Module. Figure 5 shows the connectivity diagram to install the field telephone in RAD mode.

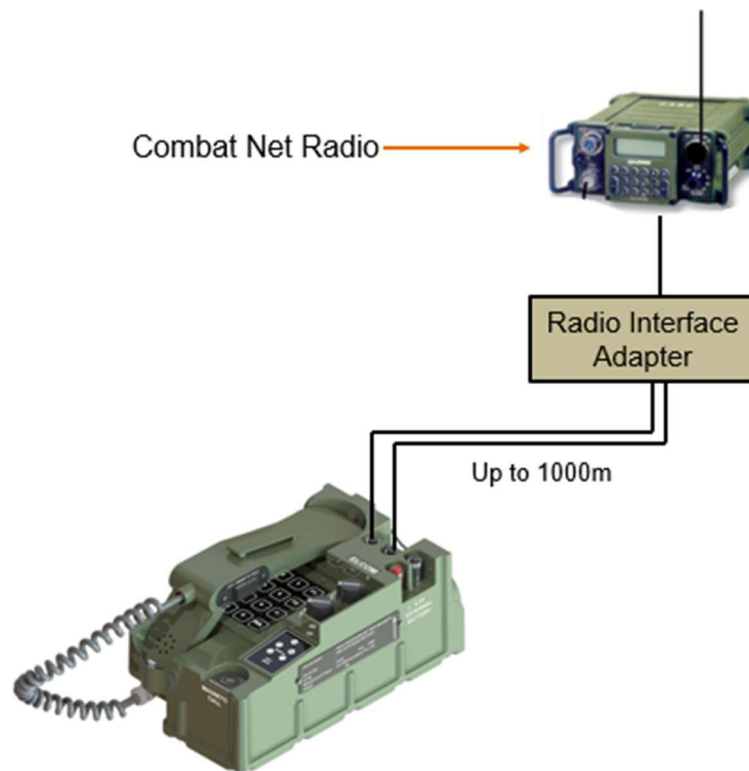


Figure 5: RFT1001 in RAD Mode

4.3 Functional Description

The Field Telephone RFT1001 is divided into a number of sub systems. Figure 6 shows the block diagram of Field Telephone Set RFT1001.

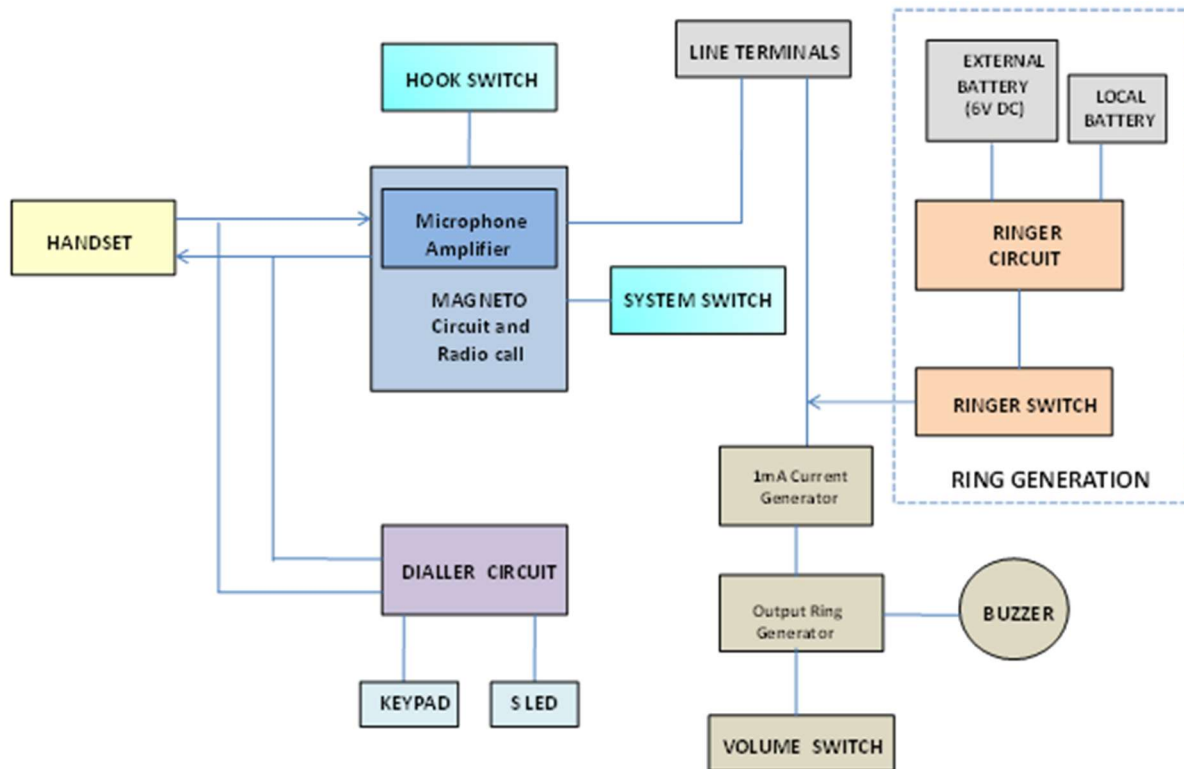


Figure 6 : Block diagram of Field Telephone Set RFT1001

A ready reference table of Field Telephone Set RFT1001 for the controls, terminals and indicators are given below.

Controls, Terminals & Indicators	Function	Location	Mode
LINE Terminals	To terminate the line wires (from CB / LB exchange or LB telephone or Radio)	Top Panel	All modes
EXTERNAL BATTERY	To connect the external battery	Top Panel	LB, RAD
'RINGER' Switch	To set ringer sound output to MUTE, LOW or HIGH	Top Panel	CBT, CRP, LB

Controls, Terminals & Indicators	Function	Location	Mode
‘SPEECH’ Switch	To set Incoming voice level to Low, Medium & High.	Top Panel	All modes
Pulse/Tone (P/T) Indicator	To show whether RFT1001 is set to DTMF or Decadic	Top Panel	CBT & CBP
‘RV’ Ring Voltage Indicator	To indicate the incoming ring	Top Panel	CBT, CBP, LB,
‘BAT MON’ key	To monitor the Battery condition (Good / Poor)	Top Panel	-
‘G’ Good Battery Status	To indicate battery status ‘Good’ when Bat. Mon. switch is pressed	Top Panel	-
‘P’ Poor Battery Status	To indicate battery status ‘Poor’ when Bat. Mon. switch is pressed	Top Panel	All modes
‘R’ Reverse Polarity of Int Batteries	To indicate that the internal dry batteries are not inserted correctly in battery box.	Top Panel	All modes
‘MAGNETO CALL’ Button	To make a call to interfaced magneto phone or to magneto exchange.	Top Panel	LB Mode
Keypad	Keypad to dial the desired numbers	Top Panel	CBT, CBP
‘SYSTEM’ Switch	To select the mode of operation - CBT, CBP, LB or RAD	Rear Panel	-
OFF/ON Hook Switch	It is used to detect the OFF-ON Hook action. It remains covered by the speaker of Handset.	Top Panel	CBT & CBP
PTT (Press to Talk) SWITCH	To activate the microphone circuit when pressed while talking.	On the Handset	All modes
Internal Battery Box	For inserting 3 nos. batteries (1.5 Volt ‘D’ Type) required for the telephone operation	Rear Wall	LB & RAD

Table 4: Control, Terminals & Indicators of RFT1001

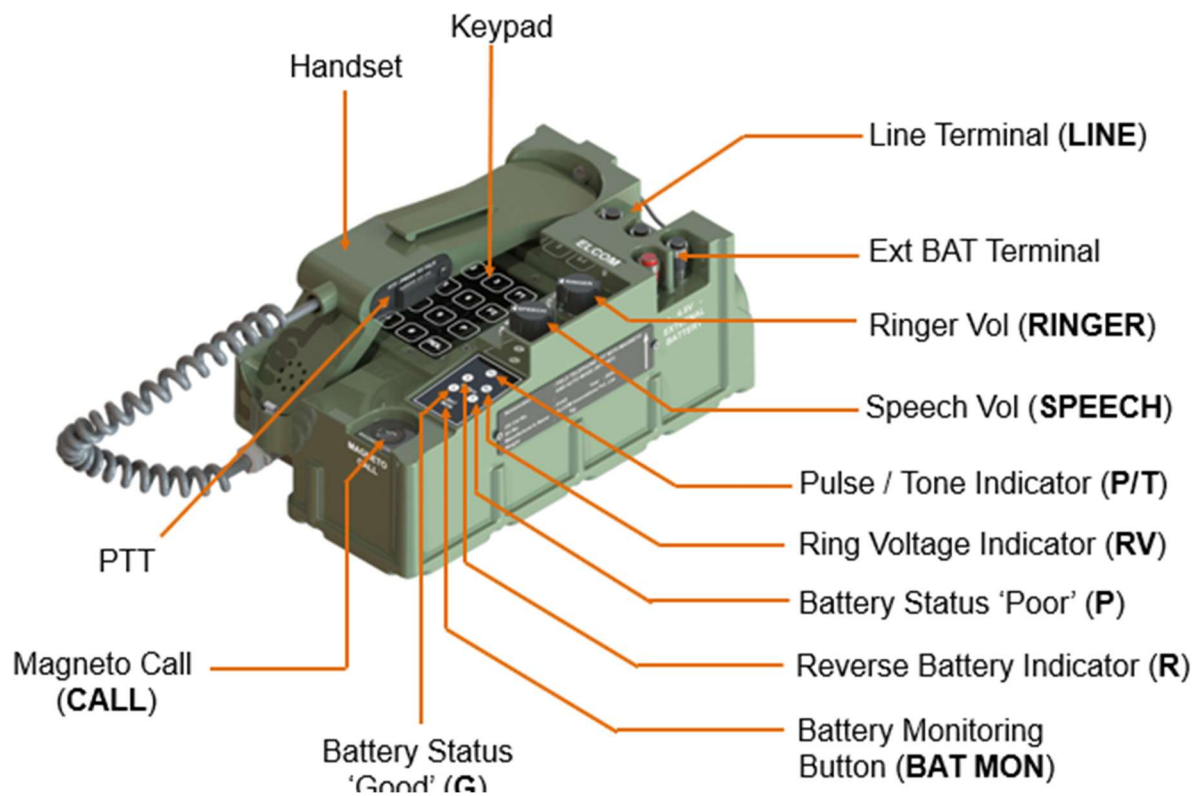


Figure 7: Controls & Terminals

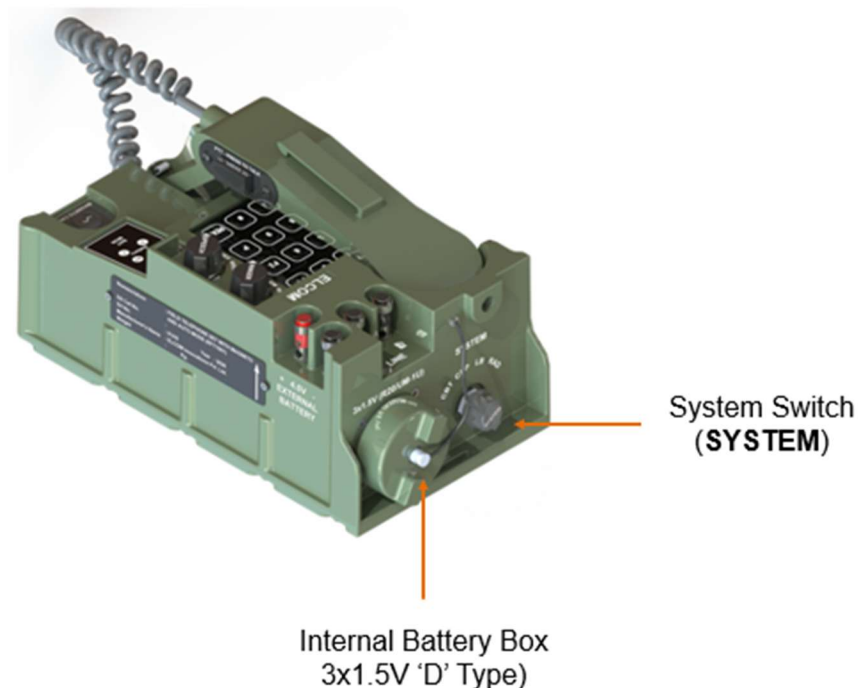


Figure 8: SYSTEM Switch & Battery Box of RFT1001

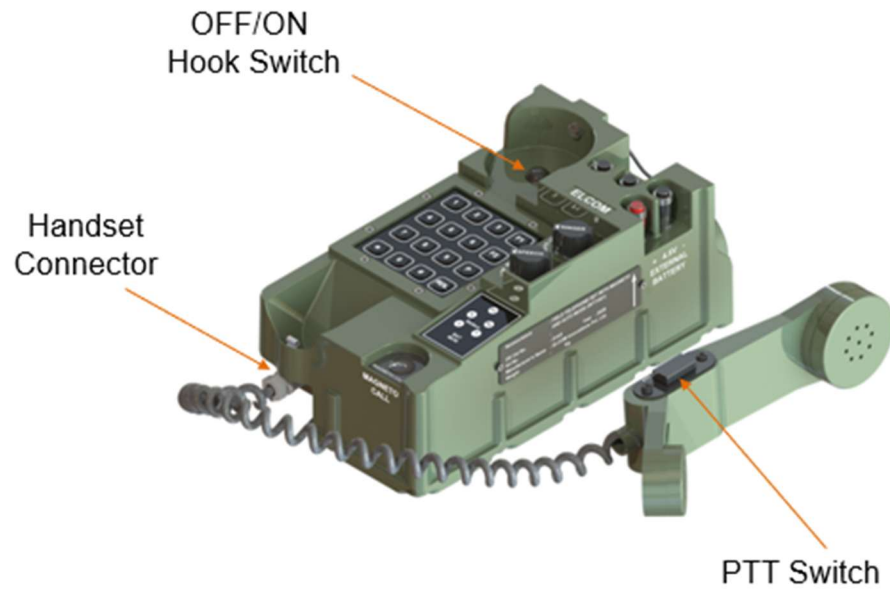


Figure 9: Hook Switch & PTT

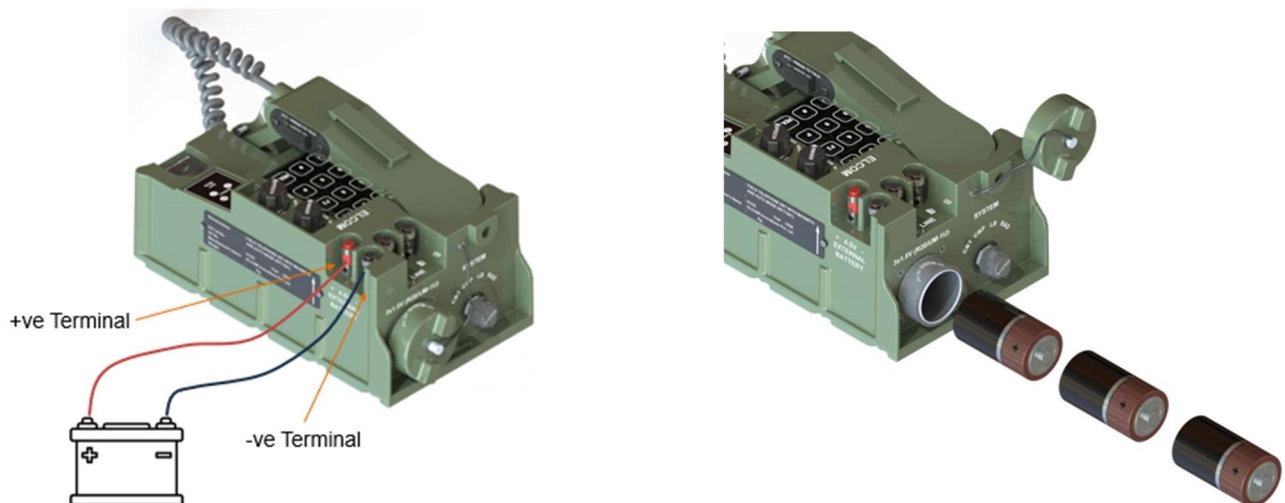


Figure 10: External & Internal Battery Connections

4.4 Description of Controls, Terminals & LEDs

4.4.1 Line Terminals

In CBT, CBP, LB and RAD mode, the field wires are to be terminated in the line terminals. Line wires will have 2 wires which can be connected to Line Terminals in any of the polarity. Figure 7 shows the terminals for terminating the line.

4.4.2 External Battery Terminals

In case of non-availability of internal batteries, field telephone RFT1001 has a provision to connect an external source (4.5 Volt DC).

The connectivity of external batteries is polarity sensitive hence the +ve & -ve wires of external battery source (4.5V) should be connected to the correct terminal of EXTERNAL BATTERY as shown in the Figure 5. The external battery source should be of 4.5 Volt.

4.4.3 Ringer Volume Switch - RINGER

The volume level of the buzzer sound can be adjusted by the RINGER switch. It is 3-position switch; MUTE, LOW & HIGH. In MUTE position, ringer will be Switched-Off.

4.4.4 Receive Volume Switch - SPEECH

RFT1001 has a incoming voice level control switch (SPEECH) to control the audio level of the handset. It is 3-position switch; LOW, MED & HIGH.

4.4.5 Pulse/Tone Indicator – P/T

This indicator shows whether the RFT1001 is set for DTMF dialing or decadic dialing. The functionality of P/T indicator will be as explained in Table 5.

Position of SYSTEM Switch	Status of P/T Indicator	Description
CBT	ON	P/T indicator will remain ON all the time
CBP	OFF	P/T will remain ON & will flicker when a number is being dialed in decadic.
LB	OFF	Not applicable.
RAD	OFF	Not applicable

Table 5: P/T Indicator

4.4.6 Ring Voltage Indicator - RV

The Incoming Ring Voltage indicator (RV) on the field telephone, shown in Figure 7, will indicate the presence of ringer voltage in the telephone line. This LED will light up when there is an incoming call to the telephone either in CBT or CBP or LB mode.

The indicator (RV) will also glow when a ringer voltage is sent down the line, while making an outgoing call in the Magneto mode (LB).

4.4.7 Battery Monitoring Key – BAT MON

RFT1001 has a BAT MON key as shown in Figure 7, which is used to view the status of the internal battery. When BAT MON key is pressed, indicators G & P are used to know the battery status (refer chapter 4.4.8).

4.4.8 Battery Status Indicators (G & P)

On pressing BAT MON key, indicators ‘G’ & ‘P’ will show the battery status. The significance of these indicators is given in Table 6.

Status of G & P	Significance	Action to be taken
For Internal Batteries		
If ‘G’ is Switched-On & ‘P’ is Switched Off.	Consolidated battery voltage is sufficient enough (i.e. > 3.5V)	No action to be taken
If ‘G’ is Switched Off & ‘P’ is Switched-On	Consolidated battery voltage is not sufficient (i.e. < 3.5V)	Batteries to be replaced
For External Power Source		
If ‘G’ is Switched-On & ‘P’ is Switched Off.	External Voltage is sufficient enough (i.e. > 3.6V)	No action to be taken
If ‘G’ is Switched Off & ‘P’ is Switched-On	Connected power source is not sufficient (i.e. < 3.6V)	Either replace the power source or increase the voltage.

Table 6: Battery Monitoring

4.4.9 Reverse Battery Indicator – R

This indicator will be switched-on if internal batteries are inserted in reverse condition in internal battery box.

4.4.10 MAGNETO CALL Button

When the field telephone is in LB mode, for making an outgoing call on the line, the MAGNETO CALL button has to be pressed for generating a ringing voltage and sending it across the line. The voltage will be generated as long as the MAGNETO CALL button is being pressed. Magneto Call Button is shown in the Figure 7.

4.4.11 Dial Keypad

The field phone has a dial keypad for dialing in DTMF and Decadic (in CBT & CBP mode), which can be used for dialing out any digits on the phone line. The keypad is a 4x4 matrix, which has several buttons as explained below.

- 0 to 9 - Dialing digits 0 to 9
- * & # - To dial * & # (*works only with CBT mode*)
- F1 - Flash key with flash time 300ms
- F2 - Flash key with flash time 640ms
- P - Pause Key with 2.2ms
- RDL - Redial Key to redial a last number dialed.

4.4.12 SYSTEM Switch

This is a 4-position rotary switch in the phone set as shown in Figure 8. The various operation modes can be set via this switch. The position is marked as shown below:

Positions marked as	Mode of operation
CBT	To set the phone to work in Central Battery mode with DTMF dialing.
CBP	To set the phone to work in Central Battery mode with decadic dialing.
LB	To set the phone in Local Battery mode
RAD	To interface the phone with radio mode through external Radio Interface Module

Table 7: SYSTEM Switch

4.4.13 OFF/ON Hook Switch

It is used to detect the OFF-ON Hook action as shown in Figure 9. It remains covered by the speaker of Handset.

4.4.14 PTT Switch

The Press-to-talk (PTT) switch is present on the handset of the field telephone. This switch is to be pressed to activate the microphone of the handset. This is applicable only in LB & RAD mode. There is no need to press PTT in CBT & CBP mode for making a conversation. PTT with handset is shown in the Figure 7 & Figure 9.

4.4.15 Internal Battery Box

The field telephone requires a bank of internal batteries (3x1.5V, 'D' type) for operating in LB and RAD mode only. This battery bank is used for generating the outgoing ringing voltage in the LB mode. For internal battery bank, 3x1.5V 'D' type batteries are to be inserted into the battery box of the field telephone as shown in the Figure 10.

5. Detail Circuit Theory

Field Telephone set RFT1001 has the following electronic boards (PCA):

- EBA30058, Control Card
- EBA30059, Ringer Card
- EBA30060, Ringer Micro Switch Card
- EBA30062, System Switch Card
- EBA30063, BAT.MON. Switch Card

The Keypad, Buzzer, Renger volume switch, Speech volume switch, Battery box, line terminals, External battery terminals and handset are directly connected to Control Card PCA through suitable connectors.

The functioning of each PCA subsystem is described in the following sub sections while Figure 11 shows the interconnection of Field Telephone Set RFT1001.

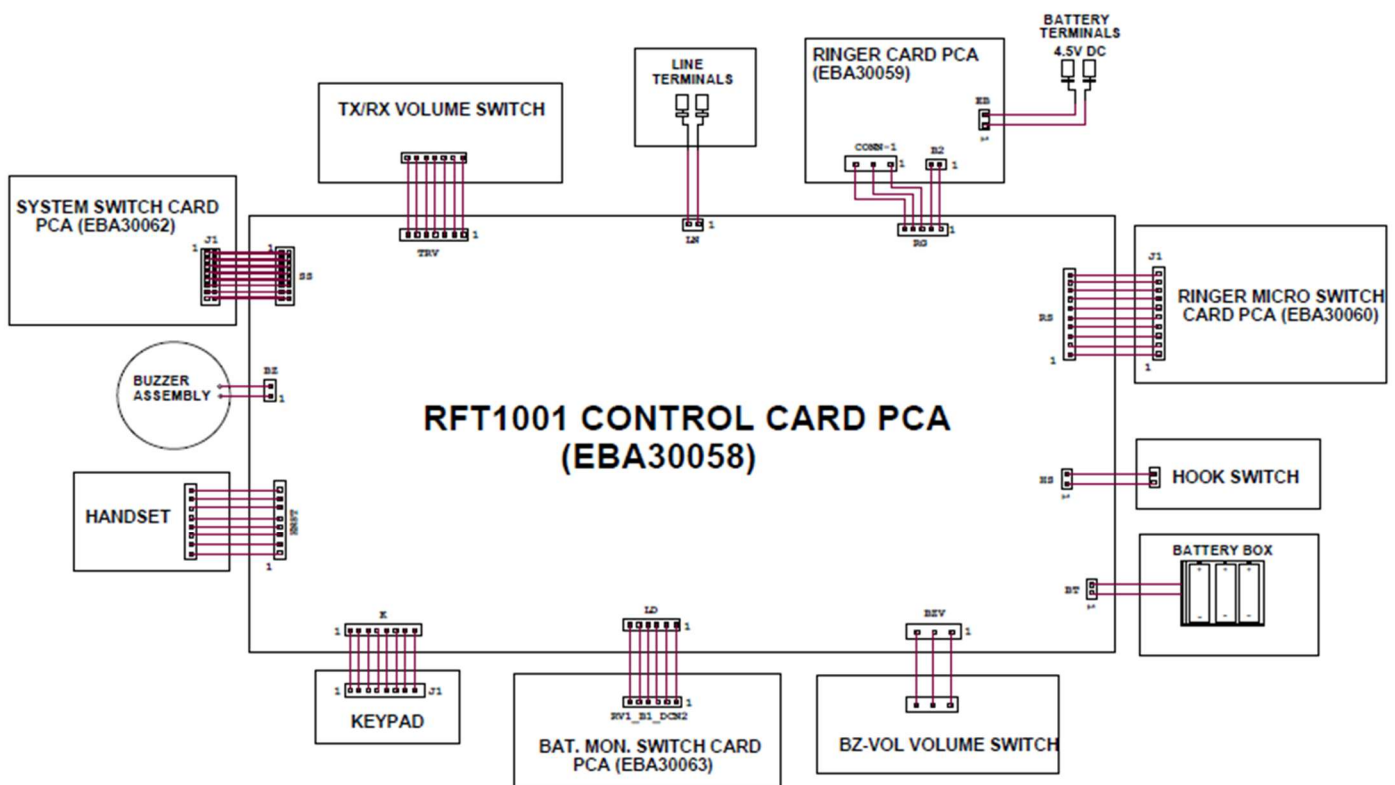


Figure 11 : Interconnection of Field Telephone Set RFT1001

5.1 EBA30058, Control Card

The block diagram of Control Card PCA is shown in Figure 12.

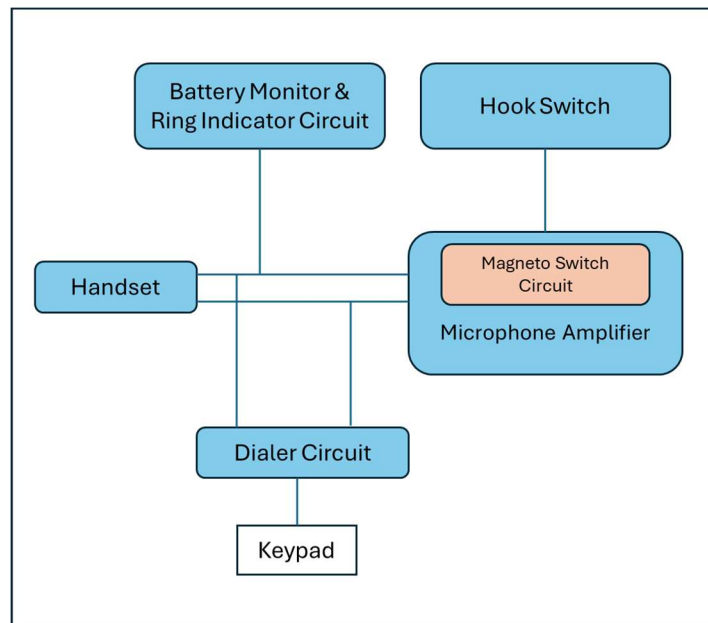


Figure 12 : Block Diagram of EBA30058, Control Card

The circuitry available on EBA30058, Control Card is mainly divided into different sections as per following details.

- a) Speech circuit
- b) Amplifier circuit for transmitting and receiving speech
- c) Dial circuit
- d) Keypad circuit
- e) Battery monitor, status & Ring Indicator circuit

Detailed description of each of these blocks is given below.

5.1.1 Speech Circuit, Line Loop Circuit & RC Coil

RFT1001 has a speech circuit which contains a handset (microphone and receiver) interface with a PTT switch, amplifier and a speech circuitry. Speech signal from the microphone is amplified by amplifier circuit using transistors. The amplified output is feed to the terminals via DC blocking circuitry. When handset's PTT presses the amplifier circuit becomes active. In ideal condition, the receiver is connected to the speech from the line thus can be heard in the receiver.

In the RAD mode, the telephone is connected to the local radio through a proprietary radio adapter. On pressing PTT of the RFT1001, the line gets looped through opto-relay and radio coil circuitry (RC coil) present in RFT1001. The line loop is detected by the radio adapter and as a result, it activates the PTT of the local radio to which it is connected. In this way, speech gets transmitted from the telephone to the radio network, on pressing the PTT. It supports sound to the buzzer upon incoming ring signal. A switch connected to it performs volume control of the output signal.

5.1.2 Amplifier Circuit

This circuit is used to amplify the Speech signal of microphone in LB mode, and the speech volume switch is in High Mode. The amplifier receives power from local battery and performs speech amplification. Positive voltage of battery is directly connected to the circuit and negative voltage is connected to the circuit when PTT of handset is pressed. When whisper mode is selected from the volume switch, the amplifier doubles the speech level.

5.1.3 Dial Circuit

RFT1001 has a Tone / Pulse dialler circuit which is used to generate standard DTMF or pulse signals of dialling as well as “flash” pulses and pause signals at line coming from central exchange in CBT / CBP mode. DTMF or pulse dialling mode can be select by changing the position of system switch in CBT / CBP Mode. In CBT / CBP mode the line voltage coming from the central exchange will appear through the dial circuit across the system switch connector causing the LED P/T to glow. During pulse dialling the same LED will flicker conforming the sending of dial pulses.

5.1.4 Keypad Circuit

RFT1001 uses a 4X4 Keypad including Flash, Pause and Redial keys. A combination of signals R1, R2, R3, R4 & C1, C2, C3, and C4 are used for each key. When any key is pressed signals goes to the dialler circuit and dialler circuit generate the signal at line coming from central exchange.

5.1.5 BAT. MON. & Ring Indicator Circuit

LED G and P: RFT1001 has a Battery Condition monitor circuit which is used to check the status (Good / Poor condition) of the battery used in RFT1001. To check the battery condition BAT MON switch is pressed. If the battery voltage is greater, then 3.5V the “G” Green LED will glow it means the batteries are in good condition. When the battery voltage goes below 3.5V the “P” Red LED will glow it means the batteries are in poor condition.

LED RV: Field telephone RFT1001 has a ring indicator LED on the top panel. This RV LED flickers when the ring signal is present on the line terminals. When the ring is generated in LB mode from RFT1001 or the incoming ring voltage is present at the line terminal in CBT / CBP mode the RV LED is flicker.

5.1.6 Connectors Detail

The Control Card PCA has several connectors, which are connected to other PCAs in RFT1001. The details of the connectors are given below:

Connector Name	Connector Type	Pin No.	Signals	Function
LINE_(LN)	2 Pin Header	1	LA	Connects line terminals from binding post PCA to Control Card PCA
		2	LB	
HOOK_SW_(HS)	2 Pin Header	1	CS1	Connects Hook switch PCA to Control Card PCA
		2	CS2	
BUZZER_(BZ)	2 Pin Header	1	BZOUT	Connects buzzer to Control Card PCA
		2	COM	
BAT_BOX_(BT)	2 Pin Header	1	BAT-	Connects Battery Box to Control Card PCA

Connector Name	Connector Type	Pin No.	Signals	Function
		2	BAT+	
HANDSET_(HNST)	8 Pin Connector	1	SPK-	Connects Control Card PCA to Handset
		2	SPK+	
		3	MIC+#	
		4	MIC-#	
		5	GND	
		6	HS6	
		7	LB/LN	
		8	RADIO	
TX/RX_VOL_(TRV)	7 Pin Header	1	SPK+	Connects Control Card PCA to SPEECH Volume Switch Assembly
		2	SPKVOL	
		3	-	
		4	-	
		5	GAS1	
		6	MED	
		7	LOW	
BZ-VOL_(BZV)	3 Pin Header	1	COM	Connects Control Card PCA to RINGER Volume Switch Assembly
		2	HI	
		3	LOW	
LED_DISPLAY_(LD)	6 Pin Header	1	N8	Connects Control Card PCA to BAT. MON. PCA
		2	LA	
		3	BAT-	
		4	BAT+	
		5	GND	
		6	VCC	
RING_PCB_(RG)	5 Pin Header	1	BAT+	Connects Control Card PCA to Ringer Card PCA
		2	BAT-	
		3	BAT+	
		4	RING2	
		5	RING1	
RING_SW_(RS)	10 Pin Header	1	HS6	Connects Control Card PCA to Ringer Micro Switch Card PCA
		2	LB	
		3	-	
		4	CS	
		5	BLN2	
		6	LA	
		7	N8	
		8	-	

Connector Name	Connector Type	Pin No.	Signals	Function
		9	RING2	
		10	RING1	
SYSTEM_SW_(SS)	16 Pin Header	1	LB/LN	Connects Control Card PCA to System Switch PCA
		2	BAT-	
		3	MAGLB	
		4	LB	
		5	MODE IN	
		6	AC1	
		7	-	
		8	AC2	
		9	GND	
		10	MAGC	
		11	QR	
		12	GND	
		13	CS2	
		14	SPKVOL	
		15	HS6	
		16	RADIO	
KEYPAD_(K)	8 Pin Header	1	R1	Connects Control Card PCA to Keypad Assembly
		2	R2	
		3	R3	
		4	R4	
		5	C1	
		6	C2	
		7	C3	
		8	GND	
		2	LINE 2	

Table 8: Control Card PCA Connector Pin Details

5.1.7 Handset Interface & PTT

This interface connects the handset / headset to the telephone. The handset / headset of RFT1001 is equipped with a PTT button. Whenever speech must be transmitted, the PTT button needs to be kept pressed. When the PTT button is released, the operator can only receive speech coming from the far end. The microphone gets switched off on releasing the PTT button. The Handset connector pin details are given below:

Handset Connector (HNST)	
Pin No.	Signal
1	SPK-
2	SPK+
3	MIC+#
4	MIC-#
5	GND
6	HS6
7	LB/LN
8	RADIO

Table 9: Handset Connector Details

PTT is used for speech transmit and #BAT- is used for providing ground for PTT sensing. TX1 and TX2 are used for speech transmit through amplifier by pressing PTT button on the handset. SPK1# and SPK2# are used for speech receive.

5.2 EBA30059, Ringer Card PCA

The ring generation circuit on the Ringer Card PCAcard receives ring generation signal through CONN1 connected to Control Card PCA. On pressing ringer switch, generates the ring voltage (90V RMS). The generated voltage is sent back through CONN1 to the main Card which extends the same to line.

The Ringer circuit receives input through EB. External batteries are used to prevent excessive use of local batteries for ring generation. The External battery voltage output connected to B2 to be sent to Control Card PCA. The block diagram of the circuit is shown in Figure 13.

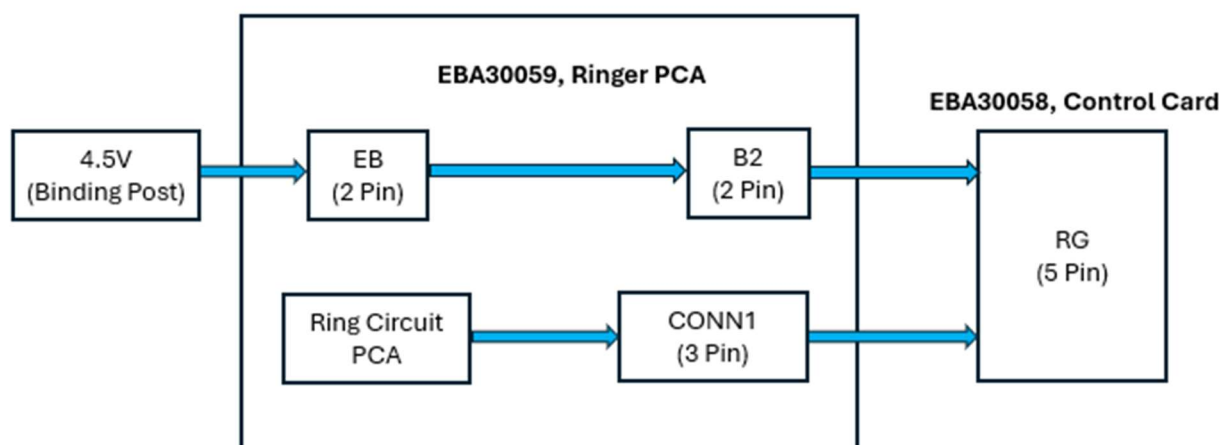


Figure 13: Block Diagram-EBA30059, Ringer Card

5.2.1 Connector Details

Connector Name	Connector Type	Pin No.	Signals	Function
CONN1	3 Pin Header	1	CONN1_1	Connects Ringer Card PCA to Main Card PCA
		2	CONN1_2	
		3	CONN1_3	
EB	2 Pin Header	1	CONN2_1	Connects Ringer Card PCA to External Battery In-Put Terminals
		2	CONN2_2	
B2	2 Pin Header	1	L1#	Connects Ringer Card PCA to Main Card PCA
		2	L2#	

Table 10: EBA30059 Ringer Card, Connector Pin Details

The circuitry involved in generation of ringer voltage is given below:

- Timer:** This device is a precision monolithic timing circuit that produces accurate time delays. On pressing the ringer switch, this device receives 5V as input, generates clock and sends it to ringer transformer.
- CMOS Multi-vibrator:** This is a low power mono-stable or a stable multi-vibrator. This device along with the ringer transformer generates the ringer voltage and sends it on line.

5.3 66079, Hook Switch Assembly

The block diagram of Hook Switch Assembly is shown in Figure 14.

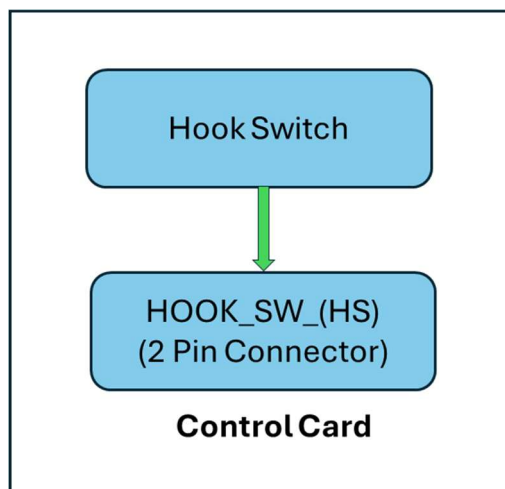


Figure 14: Block Diagram – 60079, Hook Switch Card

A 2 pole Hook switch is present on this card. Hook switch has been used for Off-hook sensing in CBT / CBP mode and used to provide high line impedance for incoming ringer in LB (magneto mode). When the handset is lifted, the Hook switch senses this and the dial tone is then engaged. For connectivity with the Main card, a 2-pin connector is present on the hook switch card.

5.3.1 Connector Details

Connector Name	Connector Type	Pin No.	Signals	Function
HOOK_SW_(HS)	2 Pin Header	1	CS1	Connects Hook switch PCA to Control Card PCA
		2	CS2	
		3	CONN1_3	

Table 11: Hook Switch Assembly - Cconnector Pin Details

5.4 EBA30060, Ringer Micro Switch Card

This Assembly has a 4PDT (Four pole double throw) Ringer switch. By pressing the ringer switch, ring voltage is generated by Ringer micro switch card PCA. This Assembly is connected to the Ringer circuit PCA using a 10-pin connector that takes the Ringer switch connections from the Ringer switch Assembly to the Control Card PCA to provide DC supply to Ringer circuit PCA for generating ring voltage and to send the generated ringer voltage from ringer circuit PCA to the Control Card PCA to send it online.

The block diagram of ringer switch assembly is shown in Figure 15.

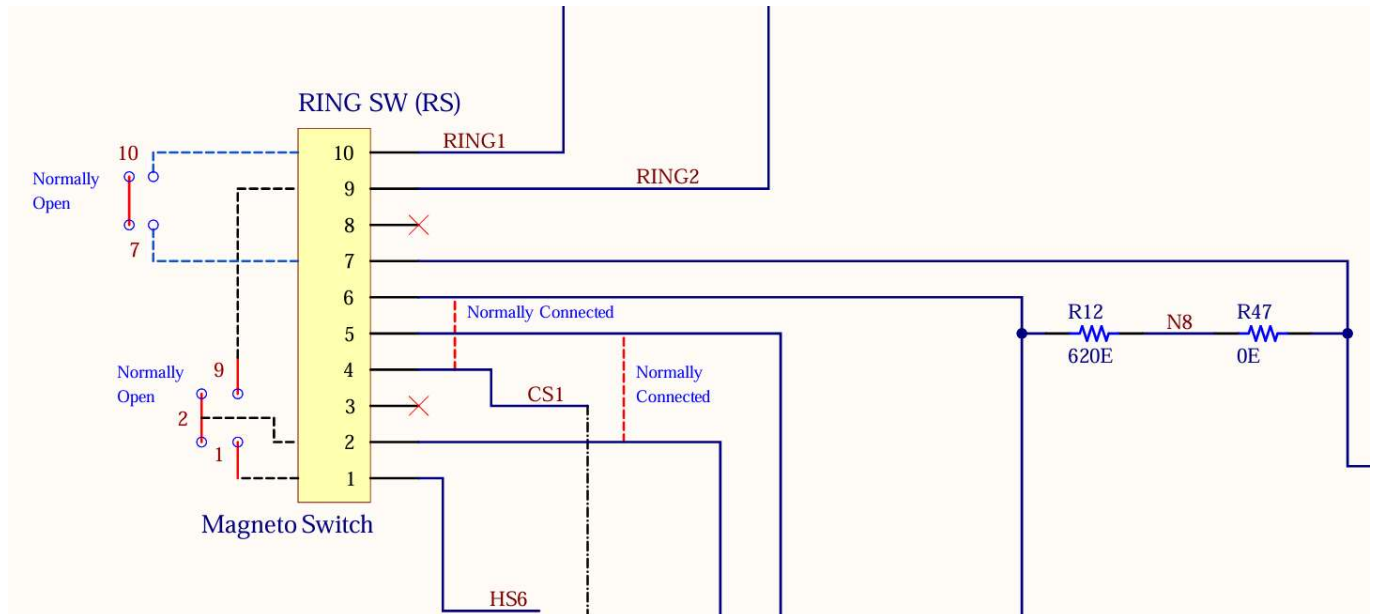


Figure 15: Block Diagram – EBA30060, Ringer Micro Switch Card

5.4.1 Connector Details

Connector Name	Connector Type	Pin No.	Signals	Function
RING_SW_(RS)	10 Pin Header	1	HS6	Connects Main Card PCA to Ringer Micro Switch Card PCA
		2	LB	
		3	-	
		4	CS	
		5	BLN2	
		6	LA	
		7	N8	
		8	-	
		9	RING2	
		10	RING1	

Table 12: Ringer Micro Switch Card PCA - Connector Pin Details

5.5 EBA30062 System Switch Card PCA

System switch card PCA is connected to the Control Card PCA by using the 16 Pin connectors. The block diagram of System Switch Assembly is shown in Figure 16.

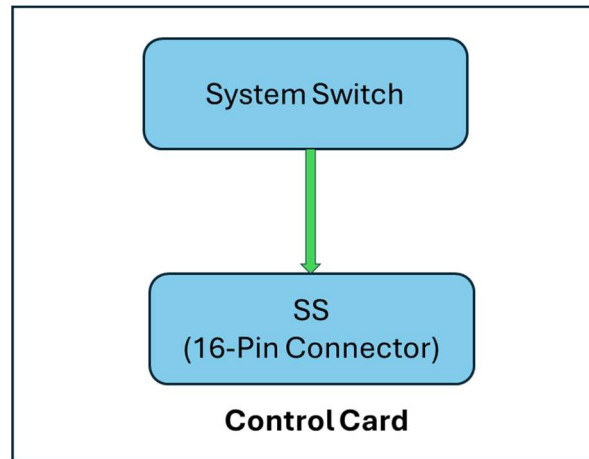


Figure 16: Block Diagram – EBA30062, System Switch Card

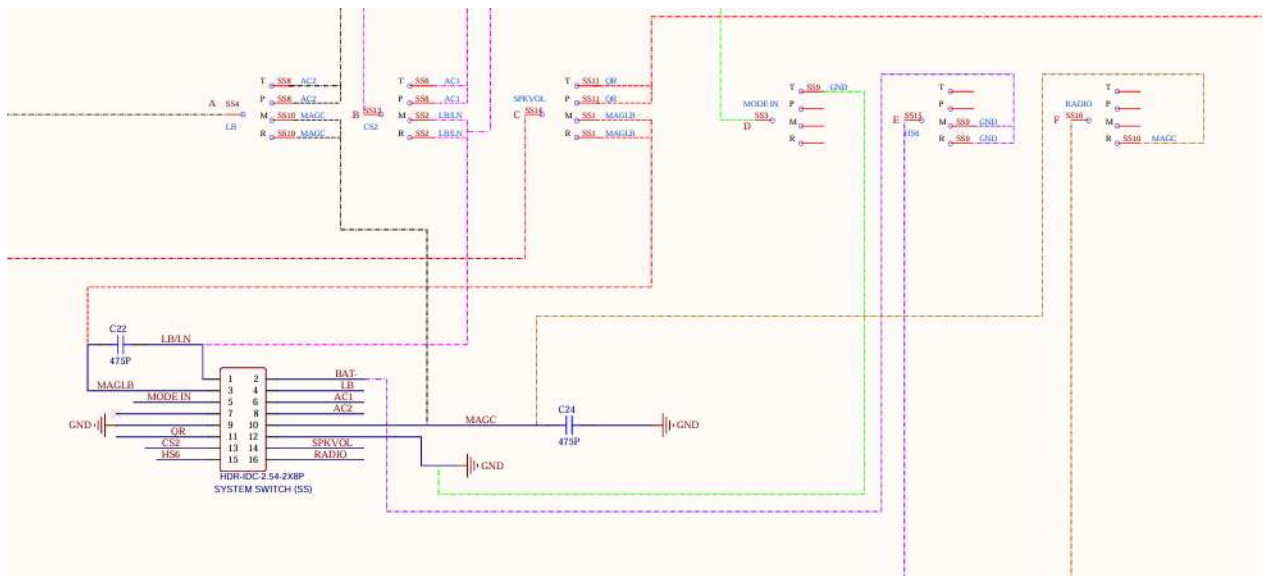


Figure 17: System Switch-Interconnection Diagram

RFT1001 has a 9 pole-4 throw switch for the system switch. System switch is used to switch the RFT1001 telephone to operate in different modes like CBT, CBP, LB and RAD mode.

- a) CBT - 1st position of the system switch is CBT in the mode the phone shall be connected to Central Battery Exchange through line terminals and keypad dialing is in Tone mode.

- b) CBT – 2nd position of the system switch is CBT in the mode the phone shell be connected to Central Battery Exchange through line terminals and keypad dialing is in Pulse mode.
- c) LB – 3rd position of the system switch is LB in the mode the phone shell be connected to another Phone or Local Battery Exchange through line terminals. keypad dialing not working in this mode.
- d) RAD – 4th position of the system switch is RAD in the mode the phone shell be connected to Radio Set through Radio adaptor or Radio Interface unit. keypad dialing not working in this mode.

5.5.1 Connector Details

Connector Name	Connector Type	Pin No.	Signals	Function
SYSTEM_SW_(SS)	16 Pin Header	1	LB/LN	Connects System Switch Card PCA to Control Card PCA
		2	BAT-	
		3	MAGLB	
		4	LB	
		5	MODE IN	
		6	AC1	
		7	-	
		8	AC2	
		9	GND	
		10	MAGC	
		11	QR	
		12	GND	
		13	CS2	
		14	SPKVOL	
		15	HS6	
		16	RADIO	

Table 13: System Switch Card - Connector Pin Details

The table below gives the pin connections of connector at various positions of the system switch.

Mode	Connector Name	Pin No.	Signals	Remarks
CBT	System_SW_(SS)	4	LB	SHORT
		8	AC2	
		13	CS2	SHORT
		6	AC1	
		14	SPKVOL	SHORT
		11	QR	

Mode	Connector Name	Pin No.	Signals	Remarks
		5	MODE IN	SHORT
		9/12	GND	
CBP	System_SW_(SS)	4	LB	SHORT
		8	AC2	
		13	CS2	SHORT
		6	AC1	
		14	SPKVOL	SHORT
		11	QR	
LB	System_SW_(SS)	4	LB	SHORT
		10	MAGC	
		13	CS2	SHORT
		1	LB/LN	
		14	SPKVOL	SHORT
		3	MAGLB	
		15	HS6	SHORT
		2	BAT-	
RAD	System_SW_(SS)	4	LB	SHORT
		10	MAGC	
		13	CS2	SHORT
		1	LB/LN	
		14	SPKVOL	SHORT
		3	MAGLB	
		15	HS6	SHORT
		2	BAT-	
		16	RADIO	SHORT
		10	MAGC	

Table 14: Signal Status in Various Positions of System Switch

5.6 EBA30063, BAT MON Switch Card

BAT. MON. Switch Card PCA is connected to the Control Card PCA by using the 6 Pin connectors. The block diagram of BAT. MON. Switch Card PCA is shown in Figure 16.

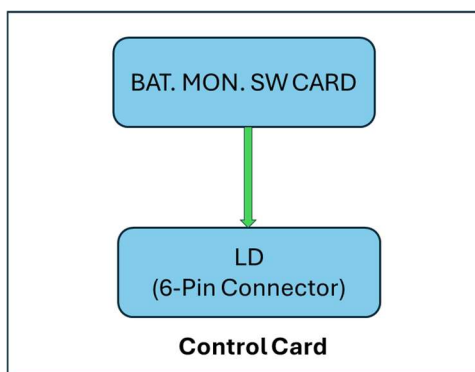


Figure 18: Block Diagram of EBA30063, BAT MON Switch Card

BAT. MON. Card has 5 LED indications and one Switch the details of the LEDs and switch as:

- P/T LED - This indicator shows whether the RFT1001 is set for DTMF dialing or decadic dialing.
- R LED – This indicator will be switched-on if internal batteries are inserted in reverse condition in internal battery box.
- RV LED – RV LED flickers when the ring signal is present on the line terminals. When the ring is generated in LB mode from RFT1001 or the incoming ring voltage is present at the line terminal in CBT / CBP mode the RV LED is flicker.
- LED G, P and BAT MON Switch - RFT1001 has a Battery Condition monitor circuit which is used to check the status (Good / Poor condition) of the battery used in RFT1001. To check the battery condition BAT MON switch is pressed. If the battery voltage is greater, then 3.5V the “G” Green LED will glow it means the batteries are in good condition. When the battery voltage goes below 3.5V the “P” Red LED will glow it means the batteries are in poor condition.

5.6.1 Connector Details

Connector Name	Connector Type	Pin No.	Signals	Function
LED_DISPLAY_(LD)	6 Pin Header	1	N8	Connects Control Card PCA to BAT. MON. PCA
		2	LA	
		3	BAT-	
		4	BAT+	
		5	GND	
		6	VCC	

Table 15: EBA30063, BAT MON Switch Card, Connector Pin Details

5.7 Ringer Volume Switch Assembly

The ringer volume switch is used to adjust the ringer volume of the telephone. The volume of ring produced can be altered by changing the modes between High, Low and Mute. RFT1001 uses a 2 pole, 3 throw volume switches. The 2nd pole is left unused. The first pole provides the High, Low and Mute modes of operation. A 3-pin harness is used to take the connections from volume switch to the Control Card PCA.

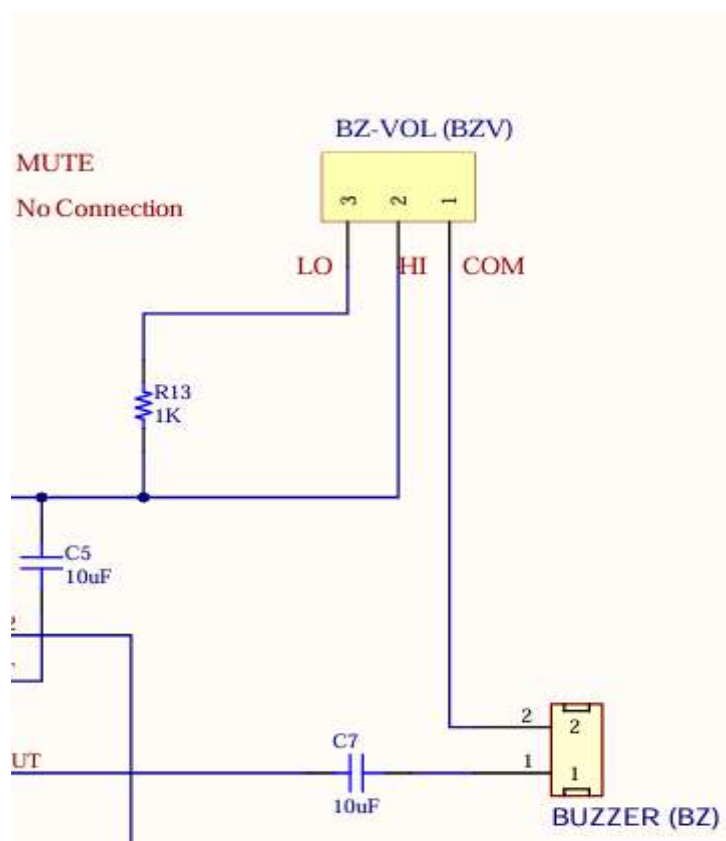


Figure 19 : Volume Switch-Interconnection Diagram

5.7.1 Connector Details

Connector Name	Connector Type	Pin No.	Signals	Function
BZ-VOL_(BZV)	3 Pin Header	1	COM	Connects Control Card PCA to RINGER Volume Switch Assembly
		2	HI	
		3	LOW	

Table 16: BZ-VOL_(BZV) Connector Pin Details

The table below gives the pin connections of connector BZ-VOL_(BZV) at various positions of the buzzer volume switch.

MODE	PIN NO. BZ-VOL_(BZV)	SIGNALS	REMARKS
MUTE	1	COM	OPEN
LOW	1	COM	SHORT
	3	LOW	
HIGH	1	COM	SHORT
	2	HI	

Table 17: Signal Status in Various Positions of BZ-Volume Switch

5.8 TX/RX_VOL_(TRV) Volume Switch Assembly

The TX/RX volume switch is used to adjust the Speech volume of the telephone. The speech volume can be altered by changing the modes between LOW, MED (medium) and High (Whisper). RFT1001 uses a 2 pole, 3 throw volume switches One pole control Speaker Volume and another works for MIC. A 7-pin harness is used to take the connections from volume switch to the Control Card PCA.

The block diagram of TX/RX volume switch Assembly is shown in Figure 19.

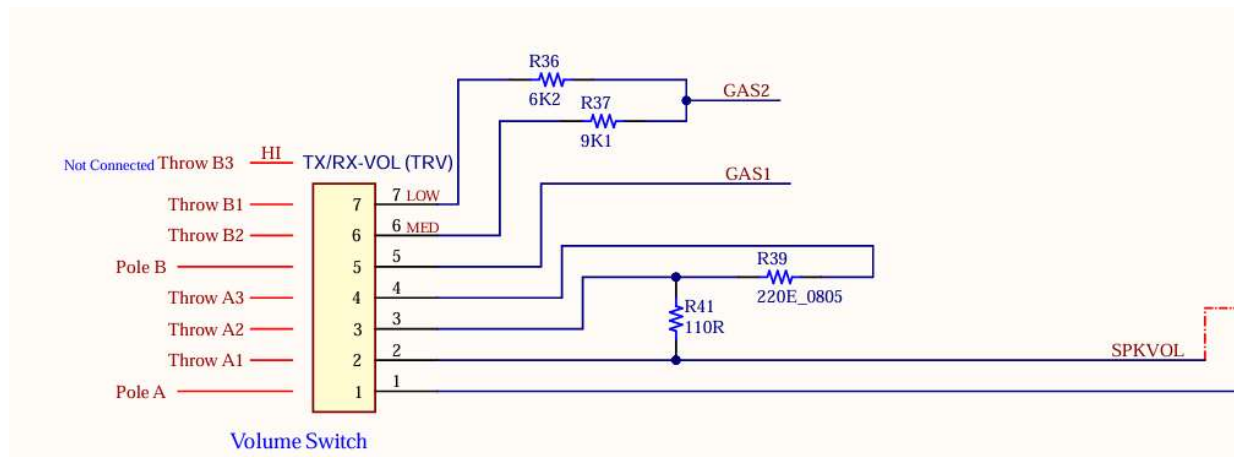


Figure 20 : TX/RX Volume Switch-Interconnection Diagram

5.8.1 Connector Details

Connector Name	Connector Type	Pin No.	Signals	Function
TX/RX_VOL_(TRV)	7 Pin Header	1	SPK+	Connects Control Card PCA to SPEECH Volume Switch Assembly
		2	SPKVOL	
		3	-	
		4	-	
		5	GAS1	
		6	MED	
		7	LOW	

Table 18: Volume switch PCA Connector Pin Details

The table below gives the pin connections of connector TX/RX_VOL_(TRV) at various positions of the Speech volume switch.

MODE	PIN NO. (J5)	SIGNALS	REMARKS
LOW	1	SPK+	SHORT
	2	SPKVOL	
	5	GAS1	SHORT
	7	LOW	
MED	1	SPK+	SHORT
	3	Resistance	
	5	GAS1	SHORT
	6	MED	
HIGH	1	SPK+	SHORT
	4	Resistance	
	5	BZ1#1	OPEN

Table 19: Signal Status in Various Positions of TX/RX VOL (TRV) Switch

5.9 Handset Connector Assembly

The block diagram of handset connector assembly detail is shown in Figure 21.

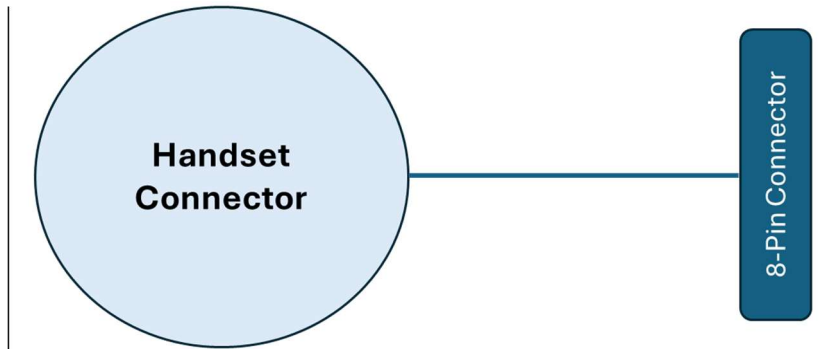


Figure 21: Block Diagram-Handset Connector

A 8-pin handset connector has been connected on this Control Card PCA through a handset harness RFT1001.

5.9.1 Connector Details

Connector Name	Connector Type	Pin No.	Signals	Function
HANDSET_(HNST)	8 Pin Connector	1	SPK-	Connects Control Card PCA to Handset
		2	SPK+	
		3	MIC+#	
		4	MIC-#	
		5	GND	
		6	HS6	
		7	LB/LN	
		8	RADIO	

Table 20 : Handset Connector Assembly Pins Detail

6. Document History

Rev.	Change	Date	Author	Approved by	
PA1-PA5	New Document	02/06/2026	SKS	AS	05/06/2026
PA6	Released	07/06/2026	SG	AS	07/06/2026

Table 21 : Document History